Exploring new states of matter in semiconductors

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Dr Alan Usher

Introduction

The study of new states of matter is of fundamental importance to physicists. We are all familiar with the three states of matter, solid, liquid and gas, which everyday elements and compounds form. However there are many other states, less common but equally fascinating. For instance the liquid-crystal state of certain compounds combines the properties of solids and liquids, and forms the basis-of electronic displays.

More exotic states can be observed when the constituent particles become lighter: liquid helium forms a super-fluid state in which all viscosity is lost; and the electrons in metals and certain ceramic materials undergo a transition to a superconducting state which conducts electricity without dissipation. It is important for us to understand these exotic states because they often provide clues to the underlying quantum mechanical behaviour of the particles involved. Extraordinary behaviour such as superconductivity also has the potential for considerable technological exploitation.

It has recently become possible to observe new states of matter in an entirely new physical system: a two-dimensional (2D) sheet of electrons. The electrons are embedded within a piece of gallium arsenide, but behave like a virtually pure system, free from imperfections. This is important in studying the transitions between various states. impurities tend to smear out such transitions (consider as an analogy the melting of butter, an impure material, compared with the melting of ice - the latter transition is much more abrupt and well-defined).

The prospect of studying new states of matter in 2D electron system is all the more exciting because electrons, unlike atoms, are fundamental, indivisible particles, one of

the simplest constituents of the universe.

We know something of the condensed states of the 2D electron system already. On lowering the temperature of the system to within one degree of absolute zero, and applying a large magnetic field, a quantum-liquid state forms, analogous to the superconducting state described above. In this state the charge carriers in the system have fractional charge (compared with the charge on an electron). At even more extreme conditions of low temperature and high magnetic field, there is good circumstantial evidence of a further transition, to a solid state in which the electrons form a triangular lattice.

Magnetisation measurements

One of the very few techniques to probe a thermodynamic equilibrium property directly, magnetisation measurements provide important new information about 2D electron systems, and potentially about their condensed states. In collaboration with co-workers at Cardiff University we have developed a unique milli-Kelvin torsion balance magnetometer with the extremely high sensitivity necessary to measure the tiny magnetisations of these sheets of electrons.

Optical studies

Photoluminescence has proved to be a useful method of examining the ground- and excited states of 2D electrons in the regimes of the quantum-liquid and electron-solid states. We are applying this technique, and that of photoluminescence-excitation, to the study of 2D HOLE systems, whose quantum condensed states are predicted to have an even more intricate and fascinating phase diagram.

See also Alan Usher's publication list.

Up to Physics Research at Exeter.
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